Multimodal RAG — Complex Test Source

Synthetic technical case study for testing PDF parsing, retrieval, multi-hop reasoning, citation precision, contradiction handling, arithmetic, and evidence grounding.

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1. Executive Summary

The Northstar Evidence Platform is a multimodal retrieval system designed to make evidence from PDFs, images, audio recordings, and video searchable. Files are stored in object storage, structured metadata is stored in PostgreSQL, and semantic vectors are stored in pgvector. A processing pipeline extracts text, speech, frames, and visual observations before creating source-scoped evidence records.

The system's objective is not merely to produce plausible answers. Each material claim must be traceable to human-readable evidence. A source is fully processed only when its required modality stages succeed. If optional visual analysis fails while OCR or transcription remains available, the source may be marked PARTIALLY_PROCESSED.

2. System Architecture

Processing consists of ingestion, extraction, enrichment, normalization, indexing, and answer generation. Ingestion records the upload and checksum. Extraction produces PDF text/OCR, audio, frames, or image OCR. Enrichment may add Whisper transcription, speaker diarization, and Vision observations. Normalization attaches timestamps, page/line metadata, and confidence. Indexing supports keyword and vector retrieval. Answering performs retrieval, reranking, verification, and grounded generation.

3. Evidence Model

Every evidence record has a globally unique database UUID and a human-readable display identifier. The immutable relationship to its source is represented by source_file_id. Evidence must never be reassigned to another source during reranking or answer generation.

An evidence record may contain: display_id, source_file_id, source_filename, evidence_type, timestamp_start, timestamp_end, page, line_start, line_end, and confidence. Example display IDs include E_VIDEO_042 and E_PDF_017.

4. Retrieval Policy

Retrieval is hybrid. Exact/keyword matching is important for identifiers and names, while vector similarity is useful for paraphrased concepts. Metadata filters are applied before final reranking when the query clearly identifies a source or modality. A query naming inspection.mp4 must not retrieve passport.png merely because both contain a semantically similar phrase.

The configured reranking formula for this test is R = 0.35S + 0.20K + 0.15M + 0.15E + 0.10C + 0.05T. S is semantic similarity, K is keyword relevance, M is metadata/source relevance, E is evidence-type relevance, C is extractor confidence, and T is temporal relevance when applicable.

5. Processing Reliability

A Vision-model timeout must not erase successful OCR or transcription. If 180 video frames are extracted and OCR succeeds on 176 frames while Vision times out after 12 attempts, the source may remain PARTIALLY_PROCESSED. The failed Vision stage is recorded separately. Repeated identical failures should be deduplicated into one error code with a count.

The configured Vision request timeout is 120 seconds. The maximum number of retries is 2, with exponential backoff delays of 2 seconds followed by 5 seconds. A worker should perform model-availability preflight instead of discovering a missing model independently for every frame.

6. Video Alignment

A video frame timestamp is measured in seconds from the beginning of the media. Transcript segments have start_time and end_time. A frame at 134.2 seconds overlaps a transcript segment from 133.7 through 136.1 seconds. Temporal overlap does not prove that the person visible in the frame is the speaker; active-speaker evidence is required for that claim.

Timeline example: 00:02:13.700 — SPEAKER_01 begins discussing the road. 00:02:14.200 — Vision records water across part of the road. 00:02:14.600 — OCR detects a roadside sign. 00:02:16.100 — SPEAKER_01 finishes the sentence.

7. Contradiction and Environment Case

field_report.pdf states on page 4, lines 18–24 that the deployment used PostgreSQL 16. deployment_notes.pdf states on page 2, lines 7–10 that the test environment used PostgreSQL 15. These statements are not necessarily contradictory because they

describe different environments.

A separate statement in field_report.pdf on page 6, lines 2–5 says that the production deployment was PostgreSQL 16. Therefore, a question specifically asking for the production database should use the page-6 evidence.

8. Numeric and Capacity Data

Daily source uploads: 2,400. Average PDF size: 3.8 MB. Average video size: 186 MB. Peak concurrent processing jobs: 48. Frame sampling rate: 2 frames/second. Embedding dimension: 768. Vector index refresh interval: 15 minutes. Target p95 retrieval latency: 850 ms.

9. Security and Sensitive Data

The platform may process identity documents. Sensitive fields must not be written to debug logs. The UI should mask identifiers by default. Retrieval must not expose a sensitive value from one source merely because an unrelated query retrieves another source. Audit logs should retain source IDs, query IDs, stage names, and error codes without unnecessarily storing full sensitive values.

10. Answer Verification Rules

A claim is SUPPORTED only when retrieved evidence directly supports it. PARTIALLY_SUPPORTED means at least one material claim is supported but another is uncertain or unsupported. UNSUPPORTED means the available evidence does not justify the requested claim.

Human-readable PDF citations use [filename — Page X, lines Y–Z]. Video citations use [filename — HH:MM:SS.mmm]. If exact PDF line numbers are unavailable, cite the page only and never invent line numbers.

11. Deliberate Edge Cases

Case A: A filename contains the same keyword as another file; retrieval must still obey explicit source references. Case B: Two sources contain different values; report the conflict and cite both. Case C: OCR has low confidence but Vision supports the reading; describe the uncertainty and use corroborating Vision evidence. Case D: A query asks who was speaking when a sign appeared; align transcript, speaker, and frame timestamps but do not equate a face with a speaker without active-speaker or identity evidence.

12. Ground Truth Statements

G1. Frame sampling rate is 2 frames/second. G2. Embedding dimension is 768. G3. Target p95 retrieval latency is 850 ms. G4. Production database in field_report.pdf is PostgreSQL 16, page 6, lines 2–5. G5. Test environment in deployment_notes.pdf is PostgreSQL 15, page 2, lines 7–10. G6. A frame at 134.2 seconds falls inside transcript interval 133.7–136.1 seconds. G7. Temporal overlap alone does not prove that the visible person is the speaker.